

Large intestine

Department of Anatomy

The abdomen (6)

Second Part

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Lecture points

- **Position of large intestine related to clinical nine regions pattern.**
- **Functions of the Large Intestine.**
- **Divisions and special features of Large Intestine.**
- **Cecum:**
 - Location and Description.
 - Cecum Relations.
 - Arterial supply.
 - Venous drainage.
 - Lymphatic drainage.
 - Ileocecal Valve.
- **Appendix**
 - ✓ Location and Description.
 - ✓ Common Positions of the Tip of the Appendix
 - ✓ Arterial supply.
 - ✓ Venous drainage.
 - ✓ Lymphatic drainage.
- **Colon:**
 - Parts.
 - **Ascending Colon.**
 - ✓ Location and Description.
 - ✓ Relation.
 - ✓ Arterial supply.
 - ✓ Venous drainage.
 - ✓ Lymphatic drainage.
 - ✓ Nerve supply.
- **Transverse Colon**
 - ✓ Location and Description.
 - ✓ Relation.
 - ✓ Arterial supply.
 - ✓ Venous drainage.
 - ✓ Lymphatic drainage.
 - ✓ Nerve supply.
- **Descending colon:**
 - ✓ Location and Description.
 - ✓ Relation.
 - ✓ Arterial supply.
 - ✓ Venous drainage.
 - ✓ Lymphatic drainage.
 - ✓ Nerve supply.
- **Sigmoid colon.**
 - Arterial supply of GUT.
 - Enteric Nervous System.

Colon

Ascending colon

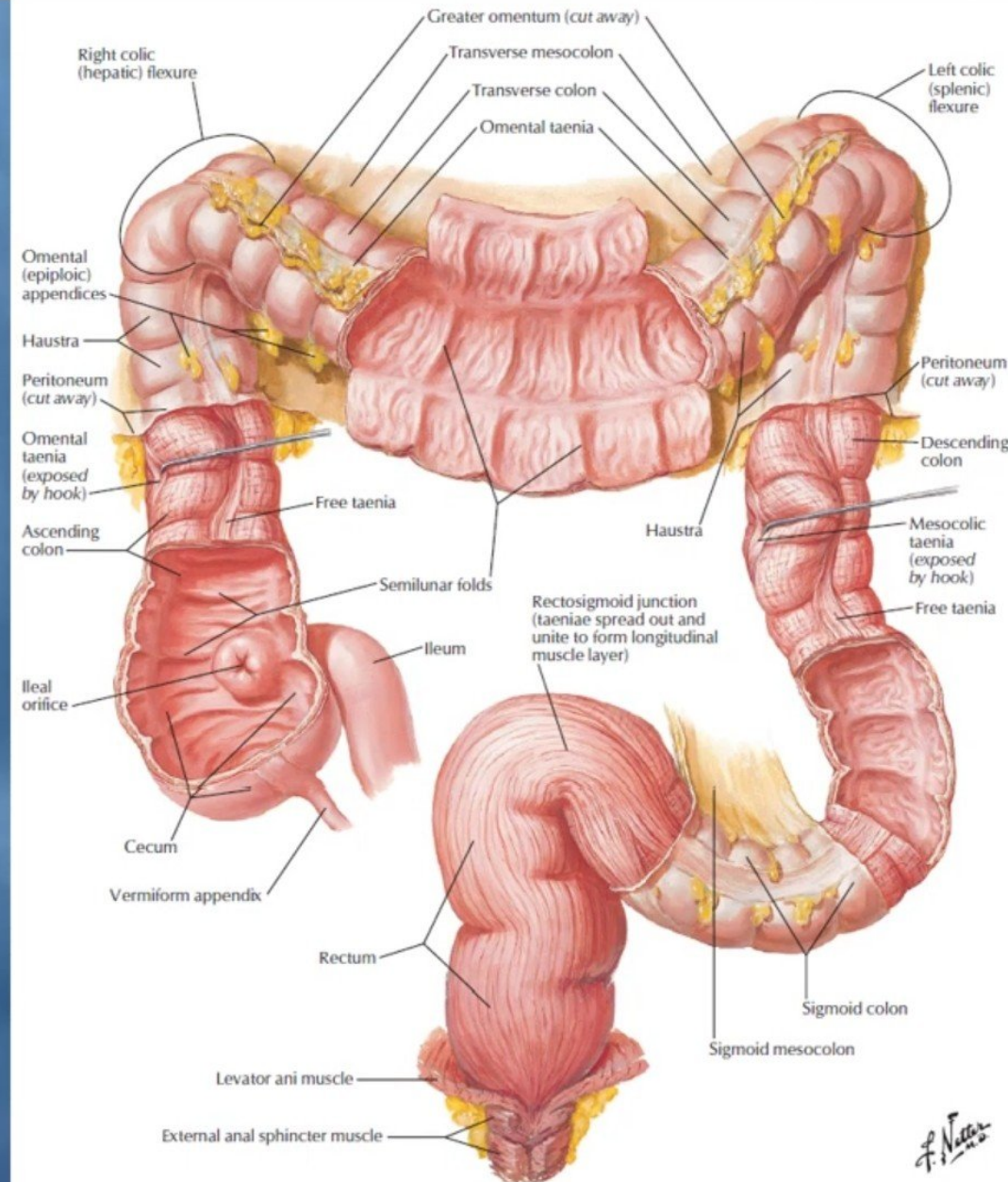
- Retroperitoneal.
- From caecum to Right lobe of liver (Hepatic flexure).

Transverse colon

- Hepatic flexure to splenic flexure.
 - ✓ Splenic flexure less mobile, anterior to inferior left kidney, attached to diaphragm via phrenicocolic ligament.
- Attached by mesocolon
 - ✓ Divides cavity into supracolic and infracolic compartments.

Descending colon

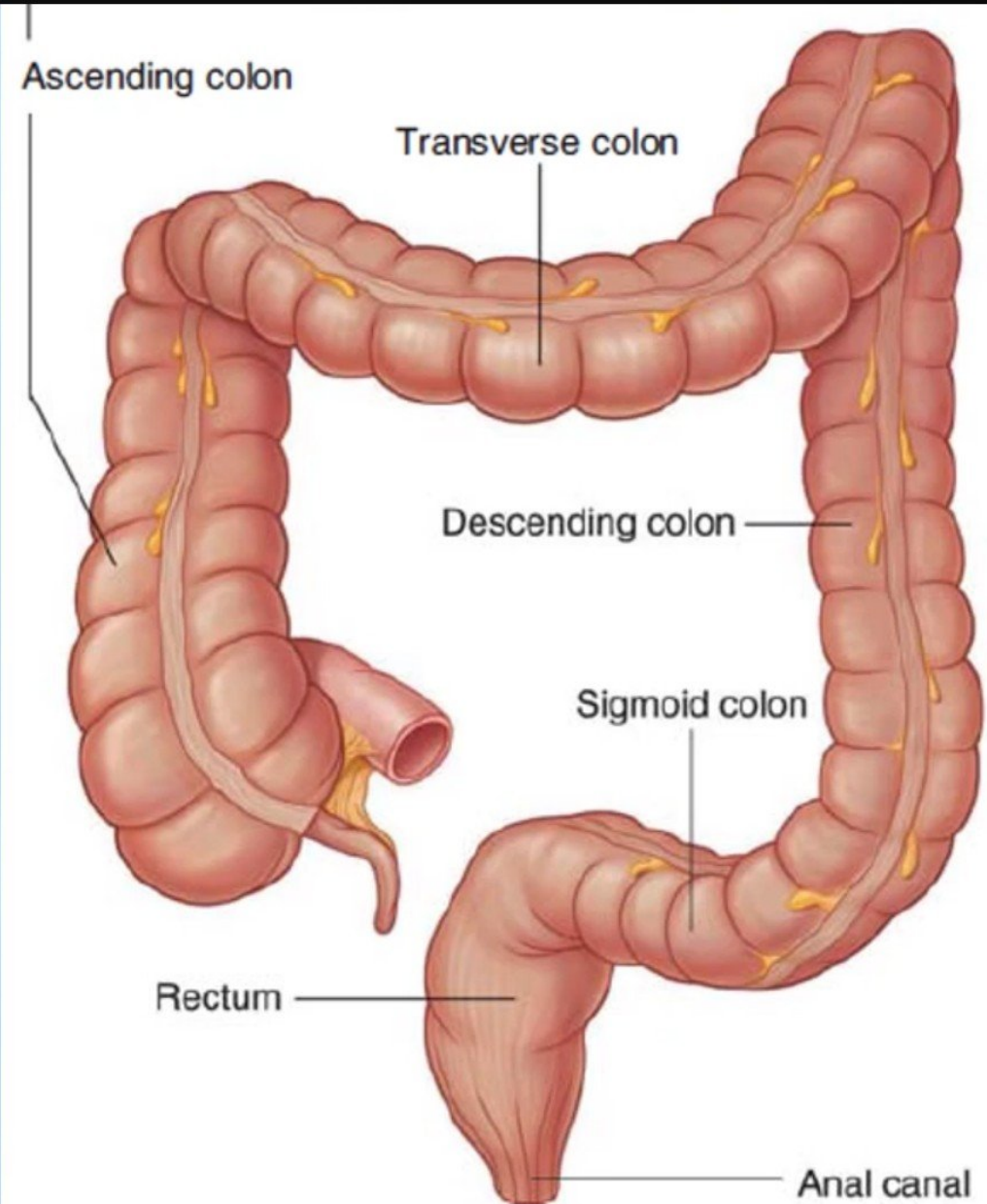
- Retroperitoneal.
- From splenic flexure to sigmoid colon.



Ascending Colon

Location and Description:

- It is about **5 in. (13 cm)** long.
- It lies in **the right lower quadrant**.
- It extends upward from **the cecum** to the **inferior surface of the right lobe of the liver**, where it turns to the left, forming the **right colic flexure**, and becomes continuous with **the transverse colon**.
- The **peritoneum** covers **the front** and the **sides of the ascending colon**, binding it to the posterior abdominal wall.



Ascending Colon

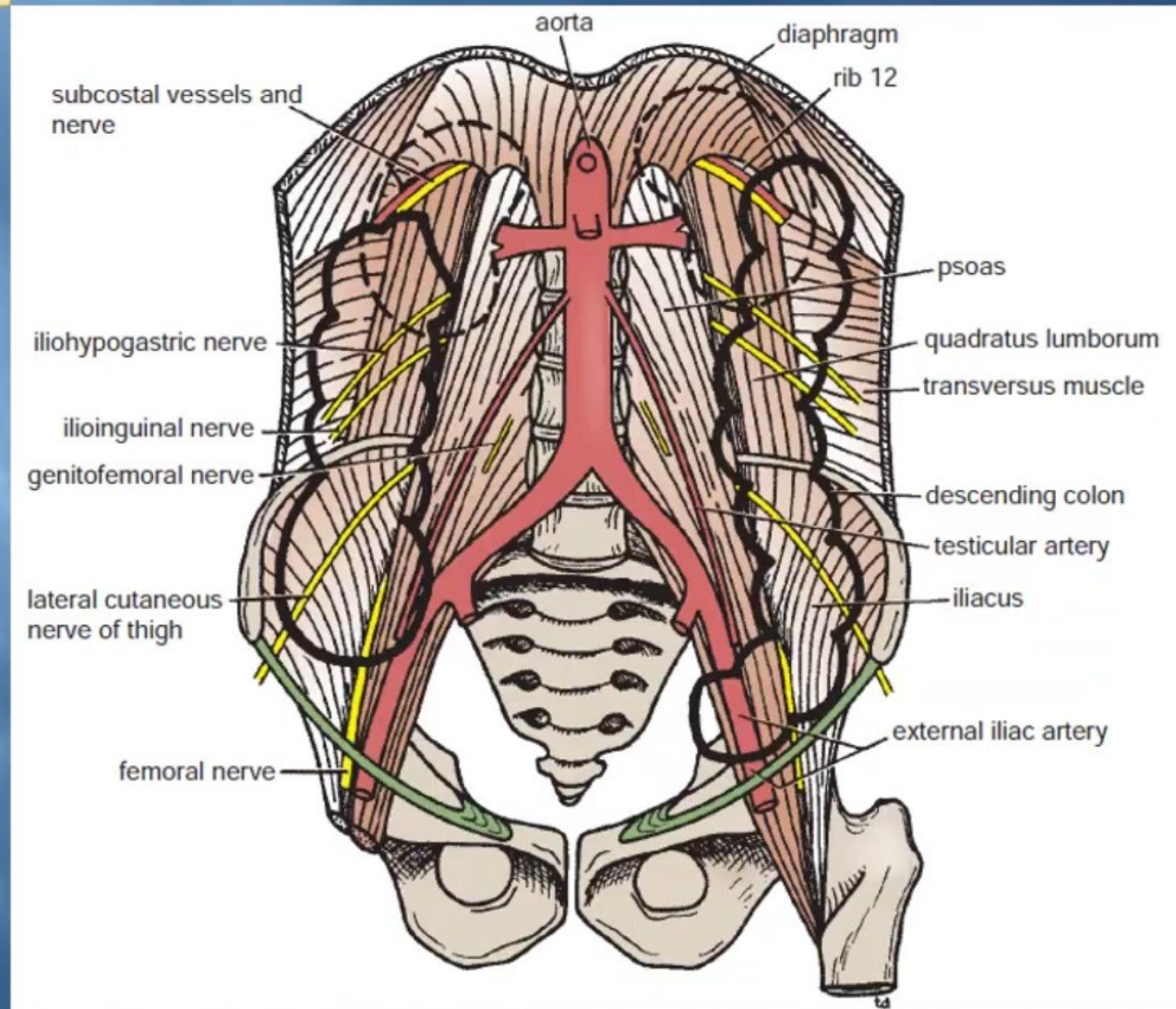
Relations

• Anteriorly:

- ✓ Coils of small intestine.
- ✓ The greater omentum.
- ✓ The anterior abdominal wall.

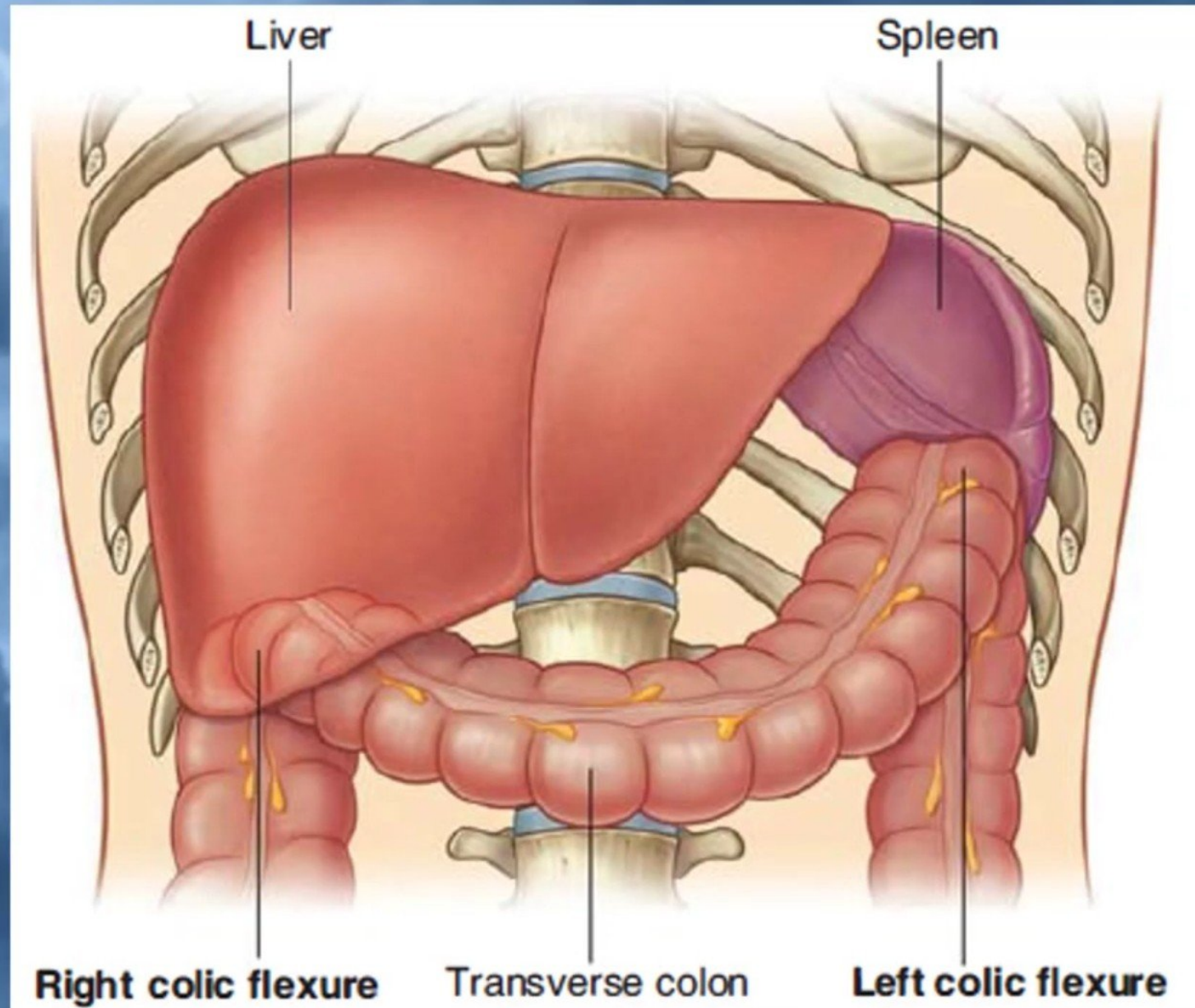
• Posteriorly:

- ✓ The iliacus muscle.
- ✓ The iliac crest muscle.
- ✓ The quadratus lumborum muscle.
- ✓ The origin of the transversus abdominis muscle.
- ✓ The lower pole of the right kidney.
- ✓ The iliohypogastric and
- ✓ The ilioinguinal nerves cross behind it.



Colon

Hepatic flexure
(right colic) and
splenic flexure
(left colic).



Ascending Colon

- Arteries:

- ✓ The **ileocolic and right colic** branches of the **superior mesenteric artery**.

- Veins:

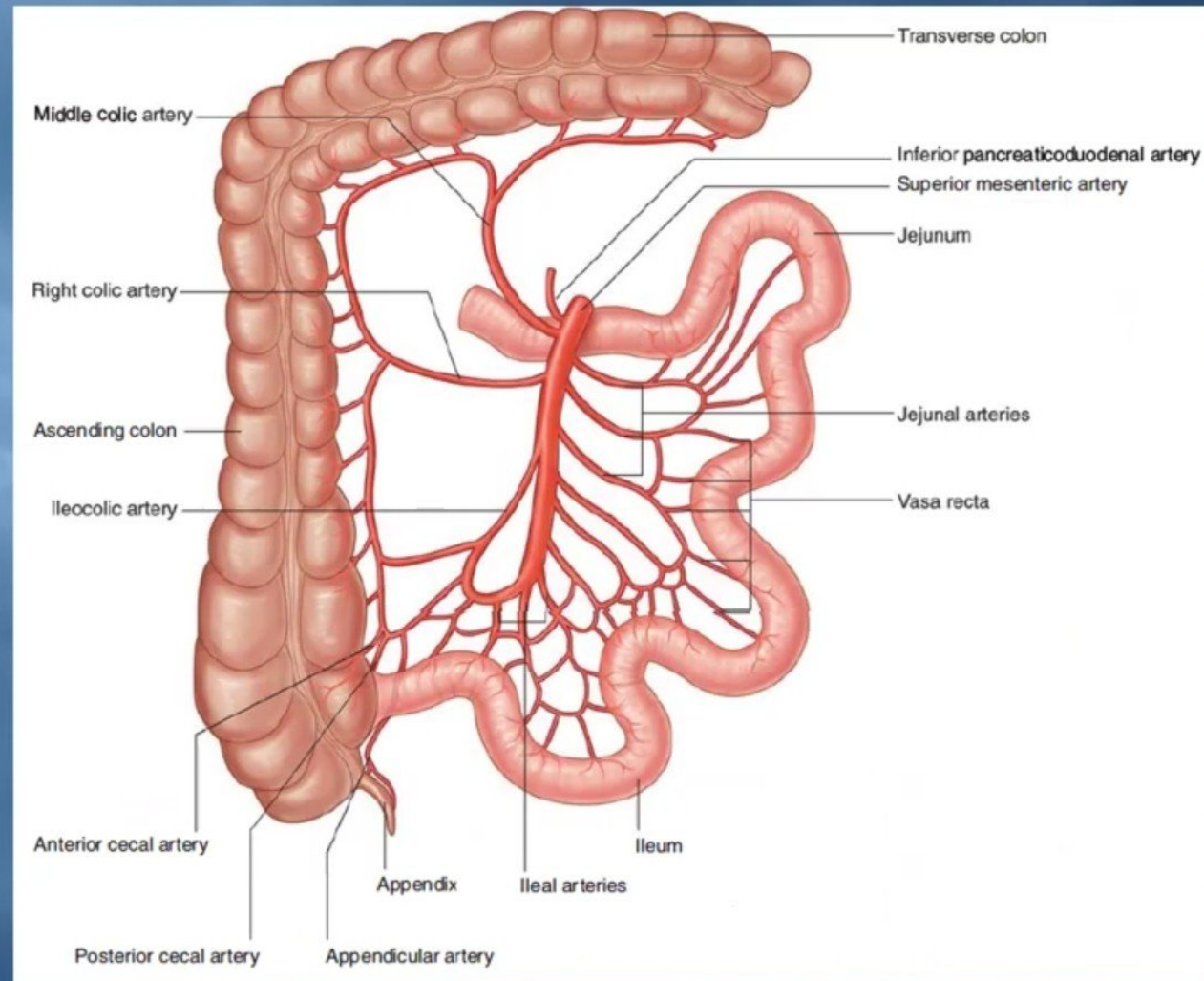
- The veins correspond to the arteries and drain into the **superior mesenteric vein**.

- Lymph Drainage:

- The lymph vessels drain into **lymph nodes** lying along the course of the colic blood vessels and ultimately reach the **superior mesenteric nodes**.

- Nerve Supply:

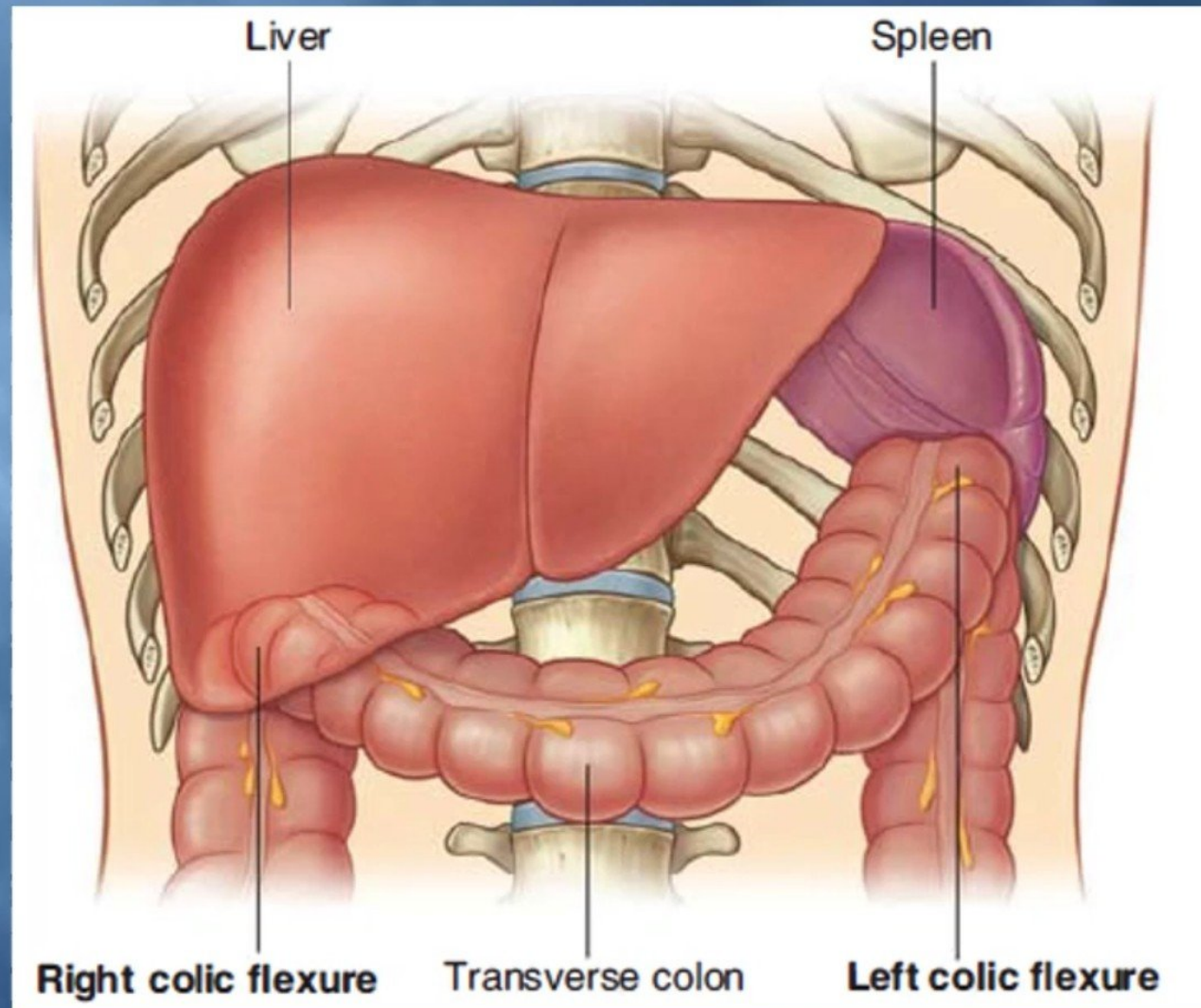
- **Sympathetic and parasympathetic (vagus) nerves from the superior mesenteric plexus** supply this area of the colon.



Transverse Colon

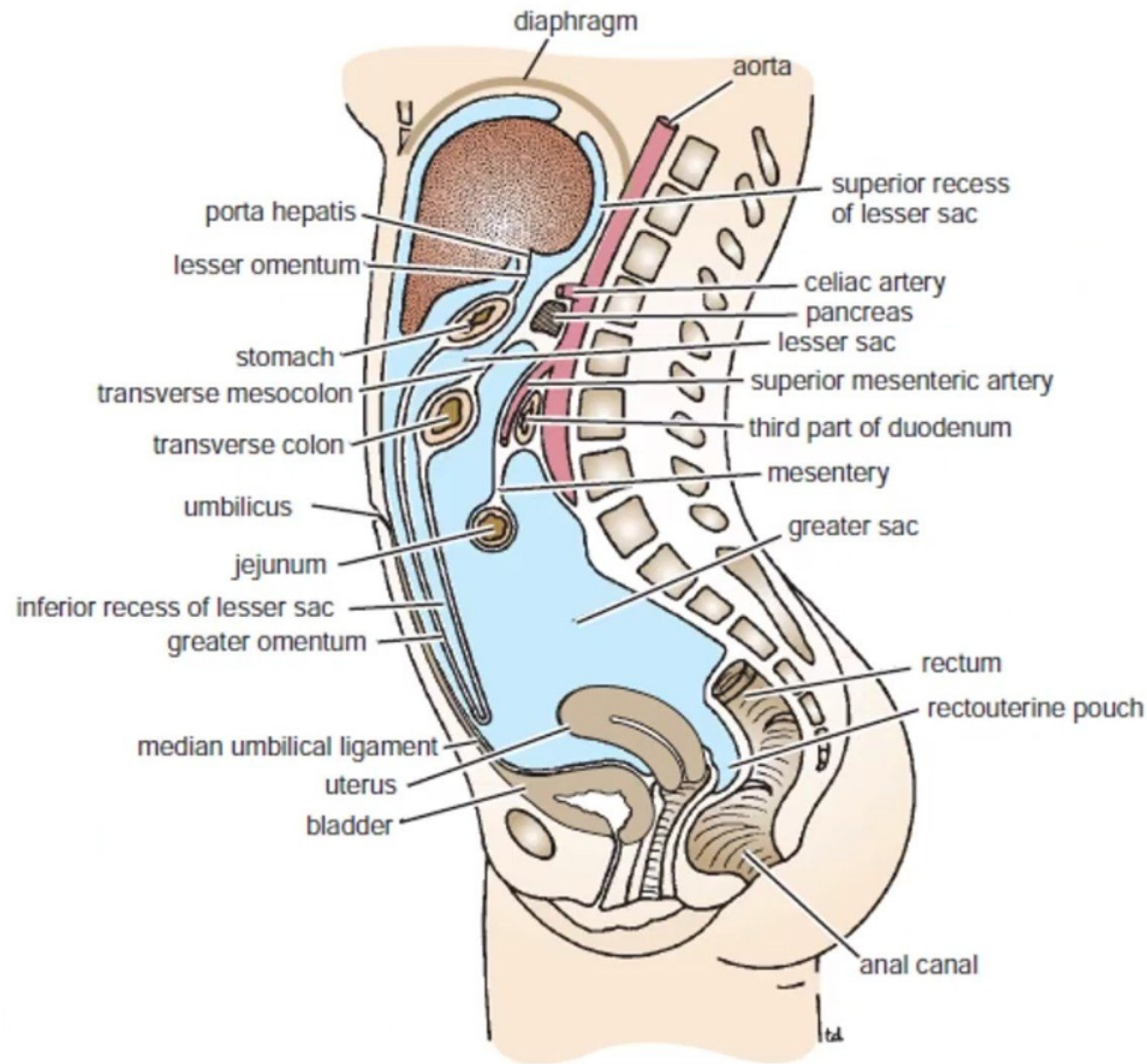
Location and Description:

- ✓ It is **about 15 in. (38 cm) long**.
- ✓ It extends across the abdomen, occupying **the umbilical region**.
- ✓ It begins at **the right colic flexure** below the **right lobe of the liver** and hangs downward, suspended by **the transverse mesocolon** from the **pancreas**.
- ✓ It then ascends to **the left colic flexure** below **the spleen**.
- ✓ The **left colic flexure** is higher than the **right colic flexure** and is suspended from the diaphragm by the **phrenicocolic ligament**.



Transverse Colon

- ✓ The **transverse mesocolon**, or (**mesentery of the transverse colon**), suspends **the transverse colon** from **the anterior border of the pancreas**.
- ✓ The **mesentery** is attached to **the superior border of the transverse colon** and
- ✓ **the posterior layers** of the **greater omentum** are attached to **the inferior border of transverse colon**.



Transverse Colon

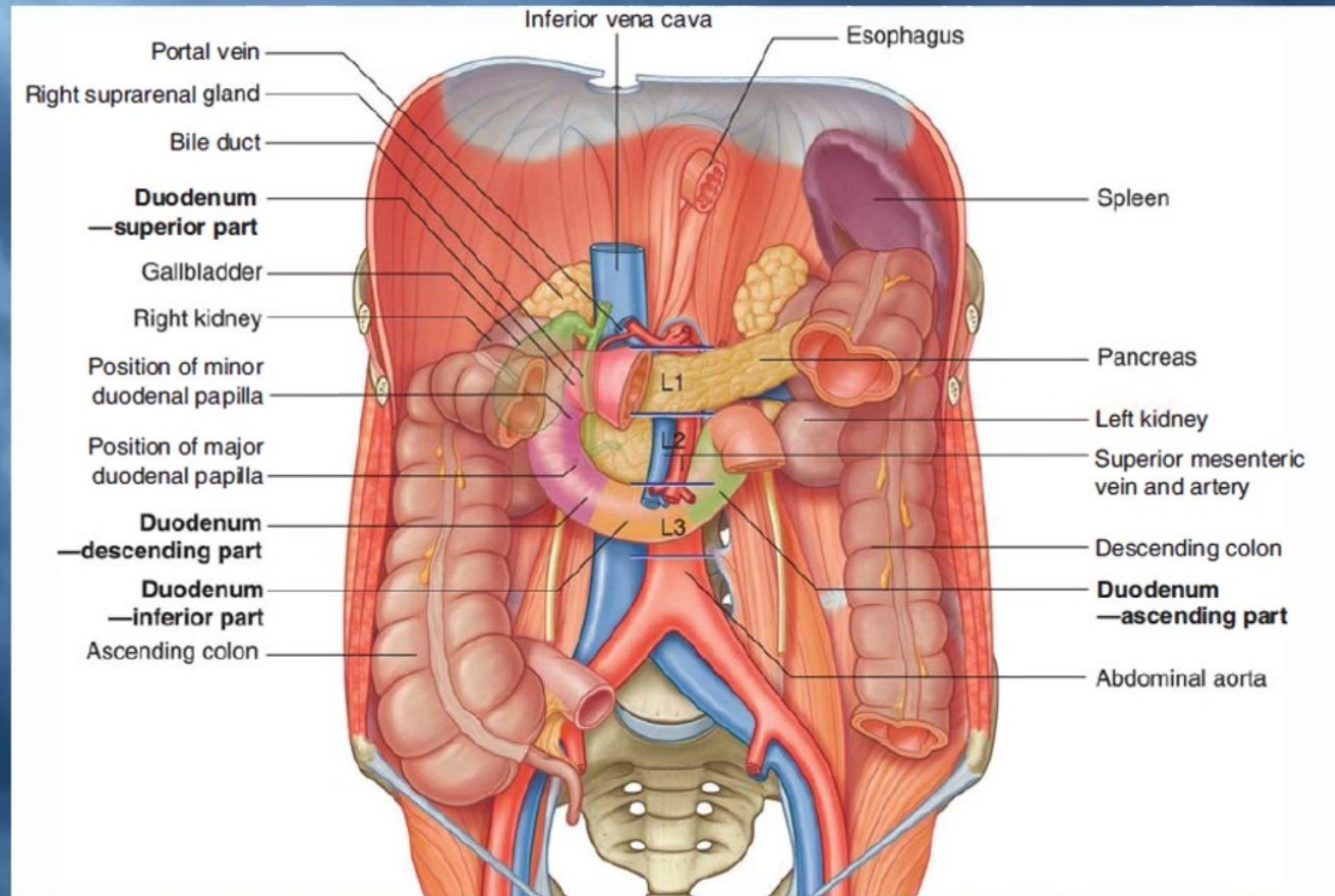
Relations:

Anteriorly:

- ✓ The greater omentum.
- ✓ the anterior abdominal wall (umbilical and hypogastric regions).

Posteriorly:

- ✓ The second part of the duodenum.
- ✓ The head of the pancreas.
- ✓ the coils of the jejunum and the ileum.



Transverse Colon

Arteries

- The **proximal two thirds** are supplied by **the middle colic artery**, a branch **of the superior mesenteric artery**.
- The **distal third** is supplied by **the left colic artery**, a branch of **the inferior mesenteric artery**.

Veins:

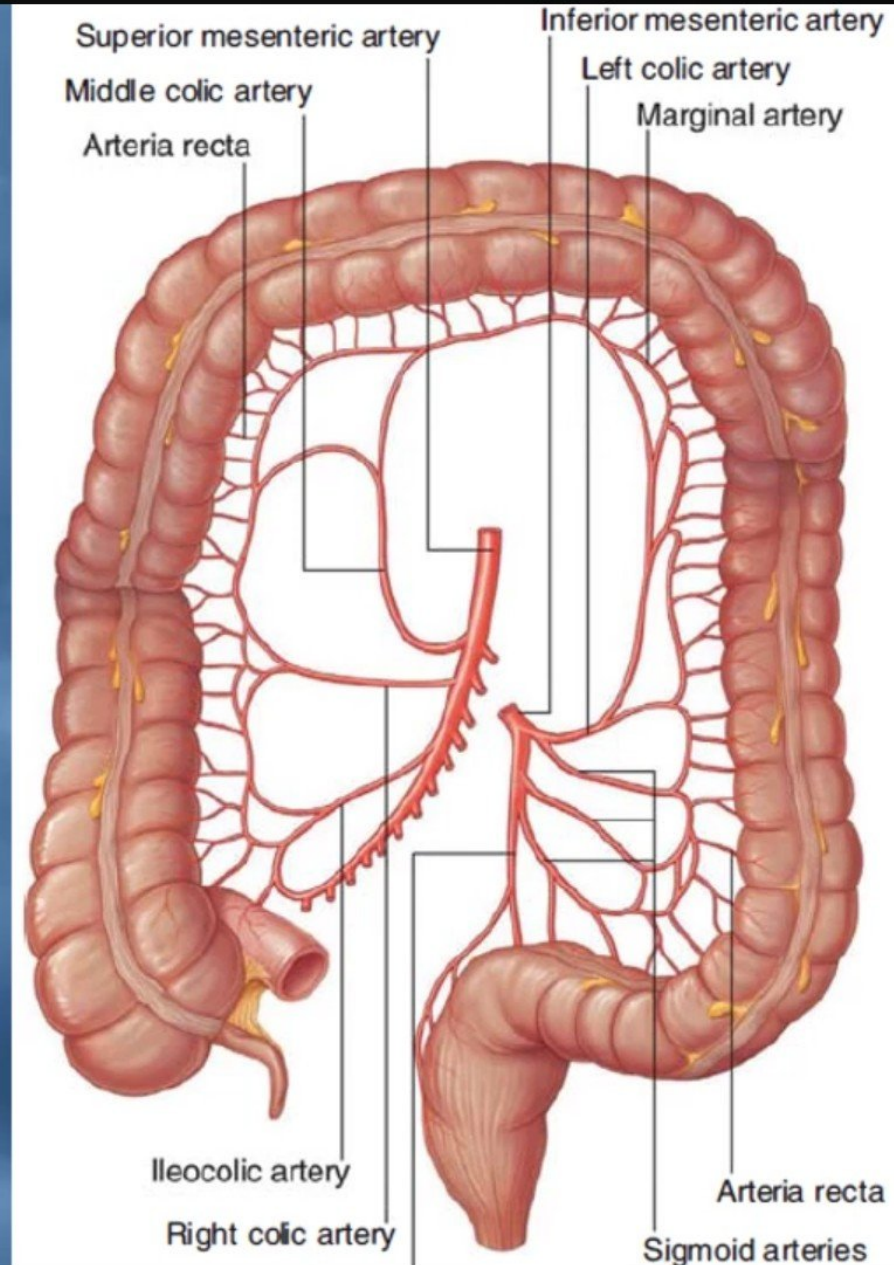
- The veins correspond to the arteries and **drain into the superior and inferior mesenteric veins**.

Lymph Drainage

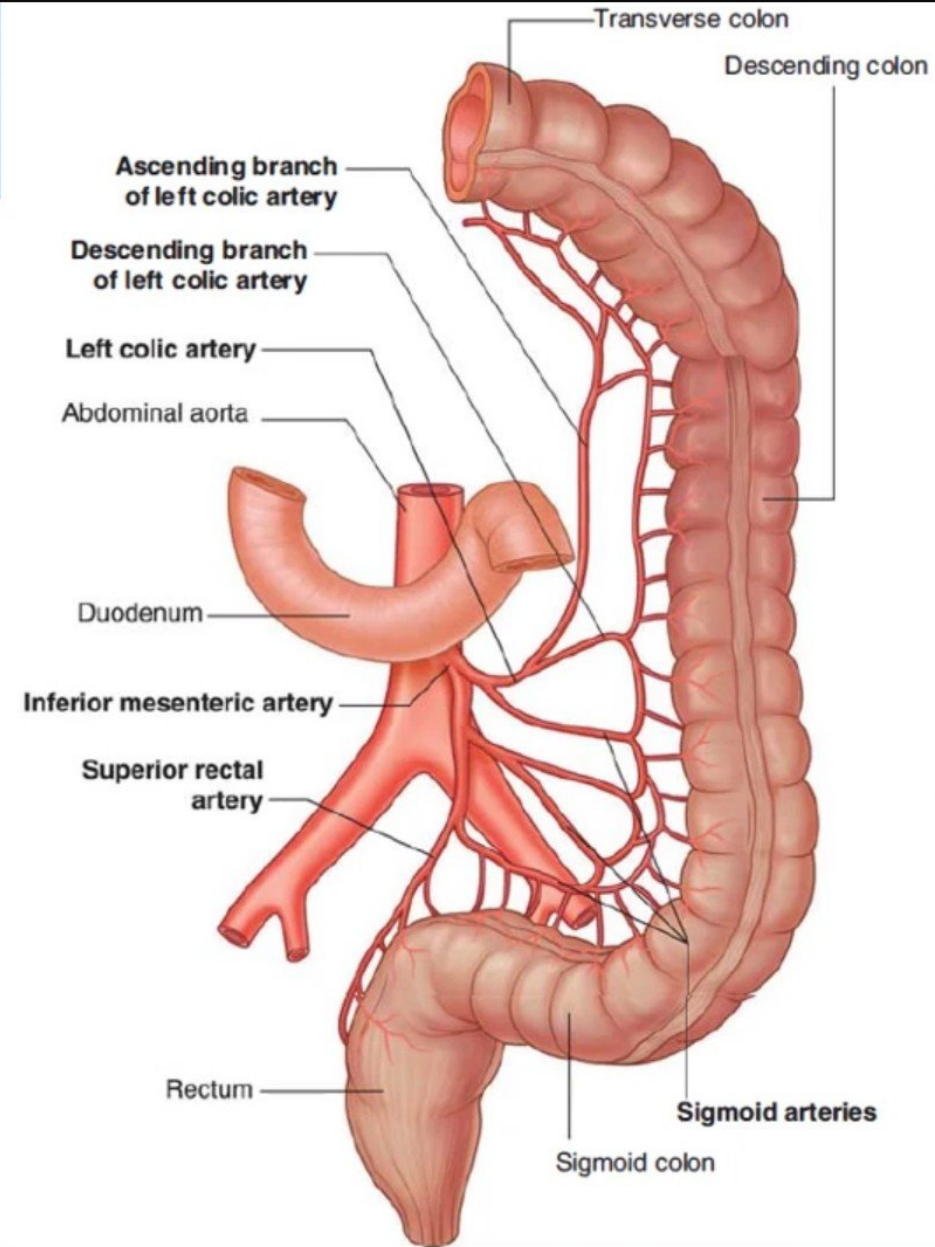
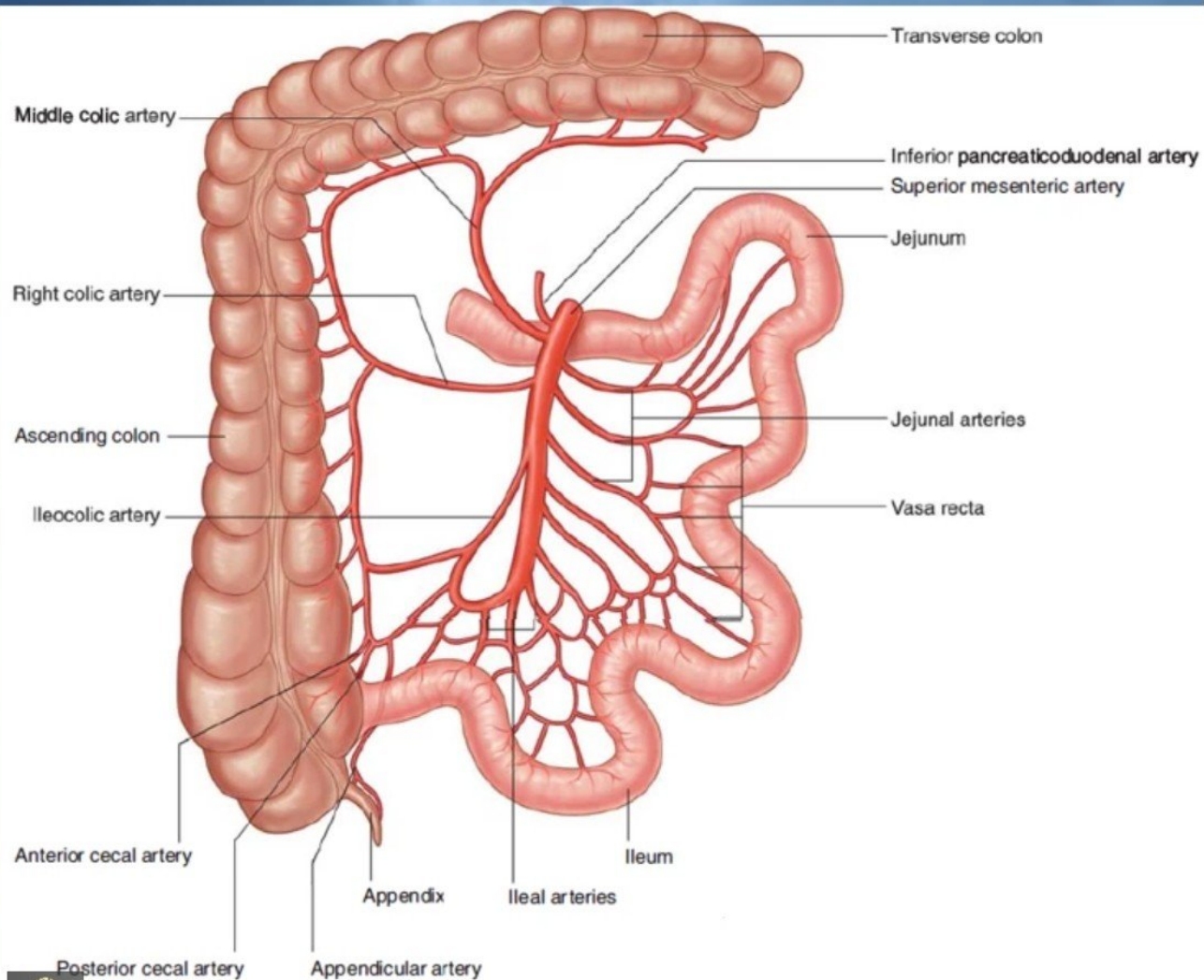
- The **proximal two thirds** drain into **the colic nodes** and then into **the superior mesenteric nodes**.
- **The distal third** drains into **the colic nodes** and then into **the inferior mesenteric nodes**.

Nerve Supply

- The **proximal two thirds** are innervated by sympathetic and vagal nerves through the superior mesenteric plexus; the distal third is innervated by sympathetic and parasympathetic pelvic splanchnic nerves through the inferior mesenteric plexus.



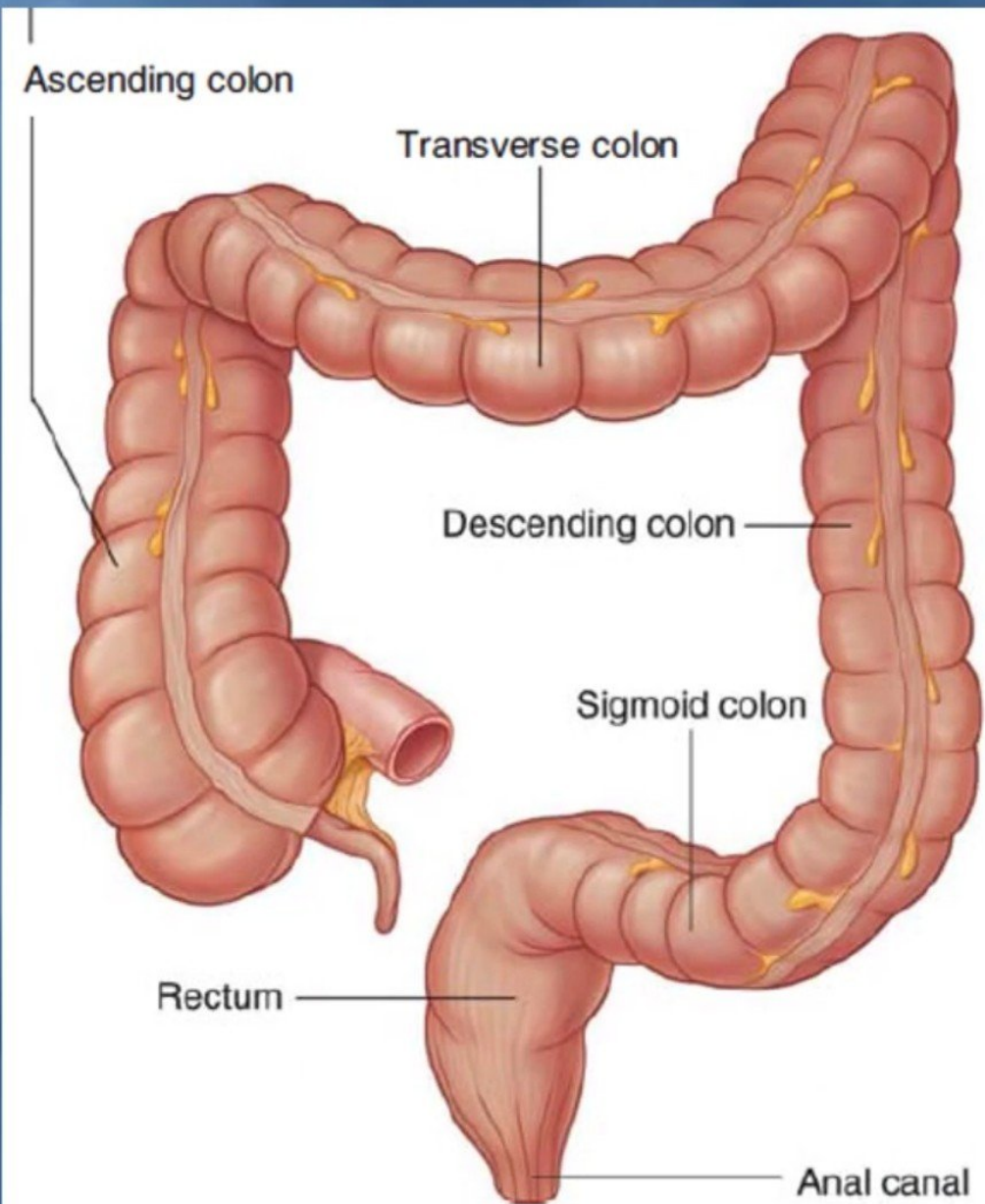
Transverse Colon



Descending Colon

Location and Description

- The descending colon is about 10 in. (25 cm) long.
- It lies in the left upper and lower quadrants.
- It extends downward from the left colic flexure, to the pelvic brim, where it becomes continuous with the sigmoid colon. (For the sigmoid colon).
- The peritoneum covers the front and the sides and binds it to the posterior abdominal wall.



Descending Colon

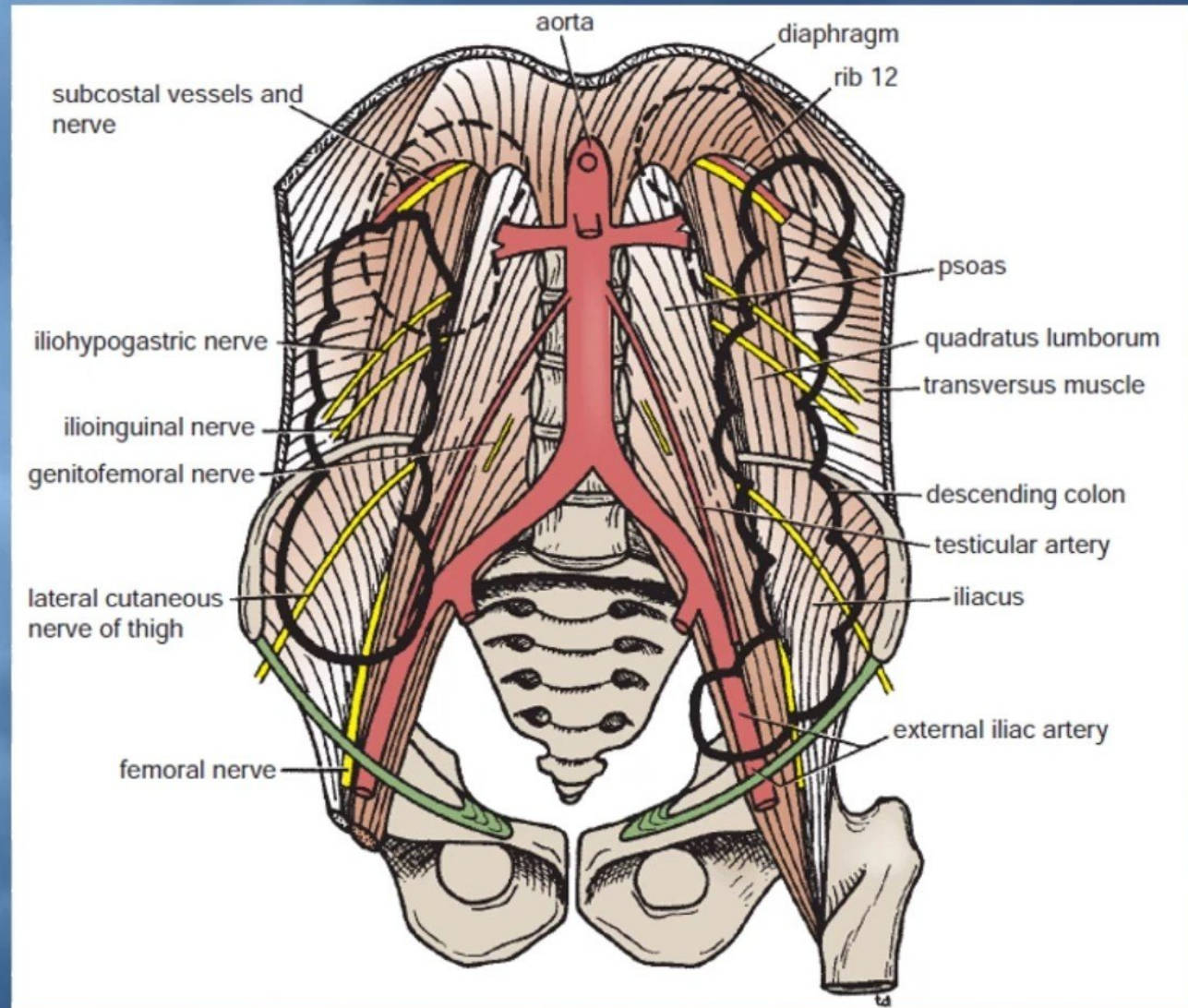
Relations

Anteriorly:

- ✓ The Coils of small intestine.
- ✓ The greater omentum.
- ✓ The anterior abdominal wall.

Posteriorly:

- ✓ The lateral border of the left kidney.
- ✓ The origin of the transversus abdominis muscle.
- ✓ The quadratus lumborum muscle.
- ✓ The iliac crest.
- ✓ The iliacus muscle.
- ✓ The left psoas muscle.
- ✓ The iliohypogastric nerve.
- ✓ The ilioinguinal nerves.
- ✓ The lateral cutaneous nerve of the thigh.
- ✓ The femoral nerve.



Descending Colon

Arteries:

The **left colic and the sigmoid branches** of the **Inferior mesenteric artery**.

Veins:

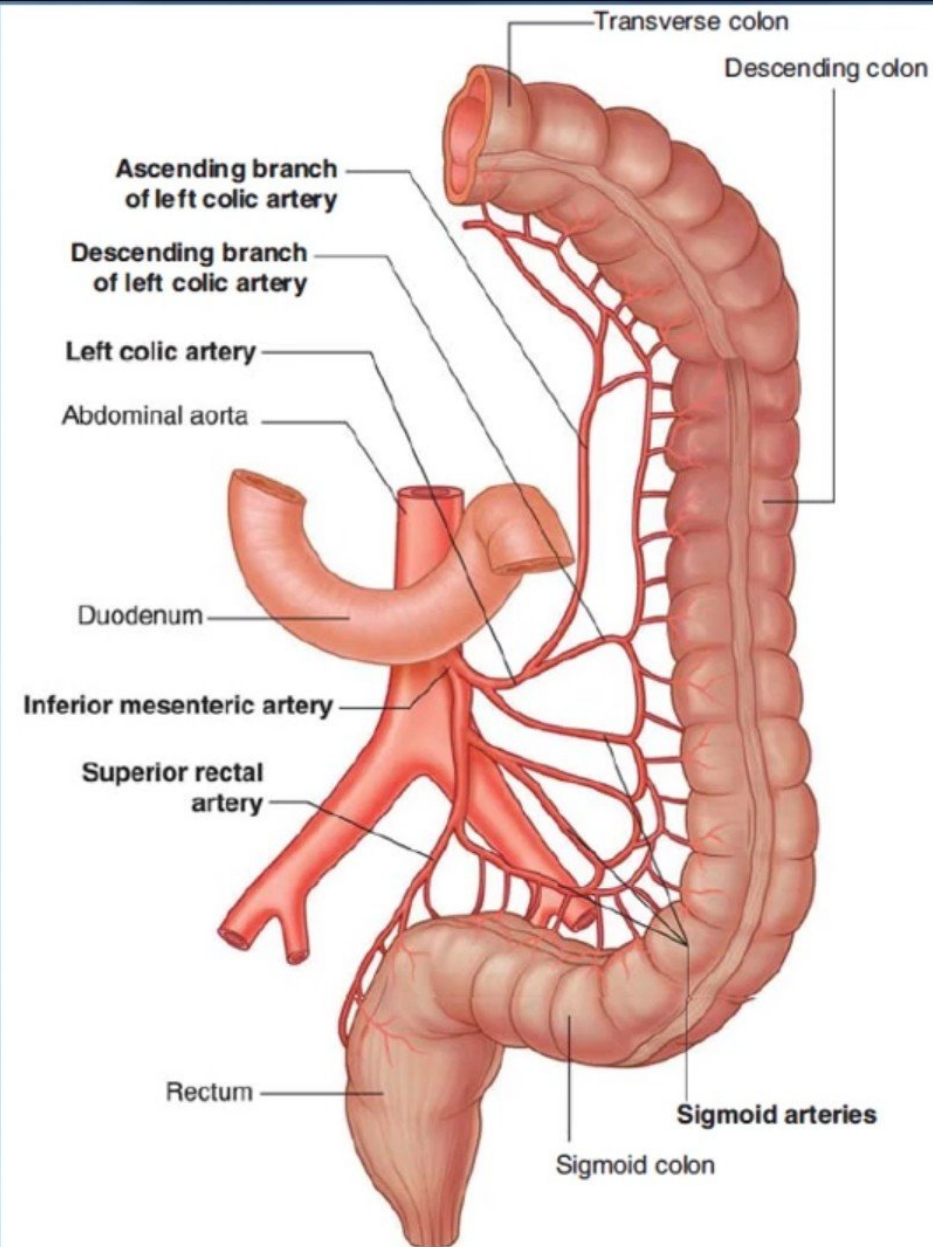
The veins correspond to the arteries and drain into **the inferior mesenteric vein**.

Lymph Drainage:

Lymph drains into **the colic lymph nodes** and **the inferior mesenteric nodes** around the **origin of the inferior mesenteric artery**.

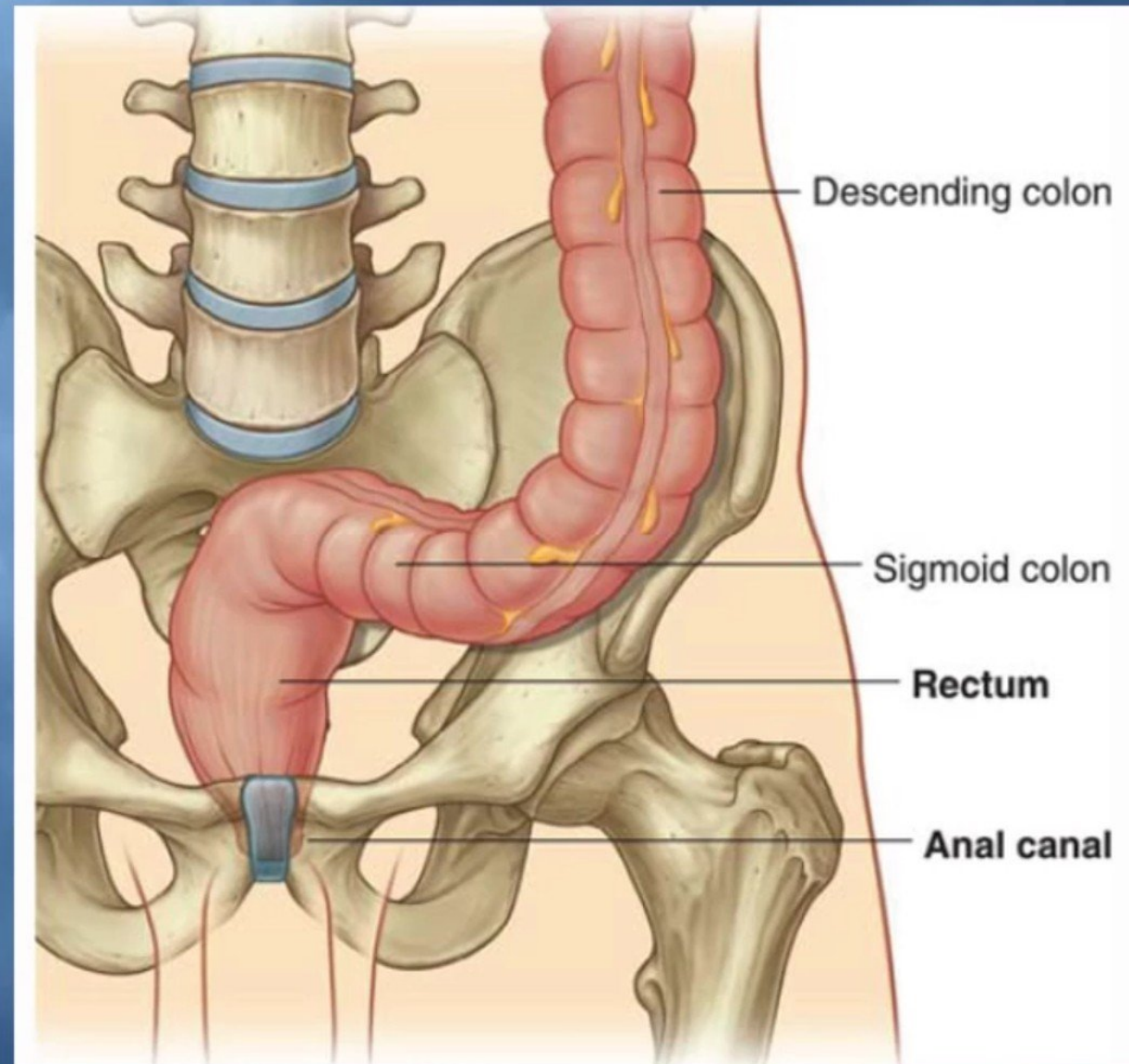
Nerve Supply:

The nerve supply is the sympathetic and parasympathetic pelvic splanchnic nerves through **the inferior mesenteric plexus**.



Sigmoid Colon

- S-shaped loop, variable length.
- It **extends** from **the descending colon** at the **pelvic brim** to **the rectum in front of the third sacrum vertebra**.
- It is the pelvic cavity structure.
- Therefore, more description in pelvic cavity region.
- It is **intraperitoneal** structure's.
- In **rectosigmoid junction** show **termination of taeniae coli**.
- **Left ureter** and **division of left common iliac artery** found **posterior to root of sigmoid**.

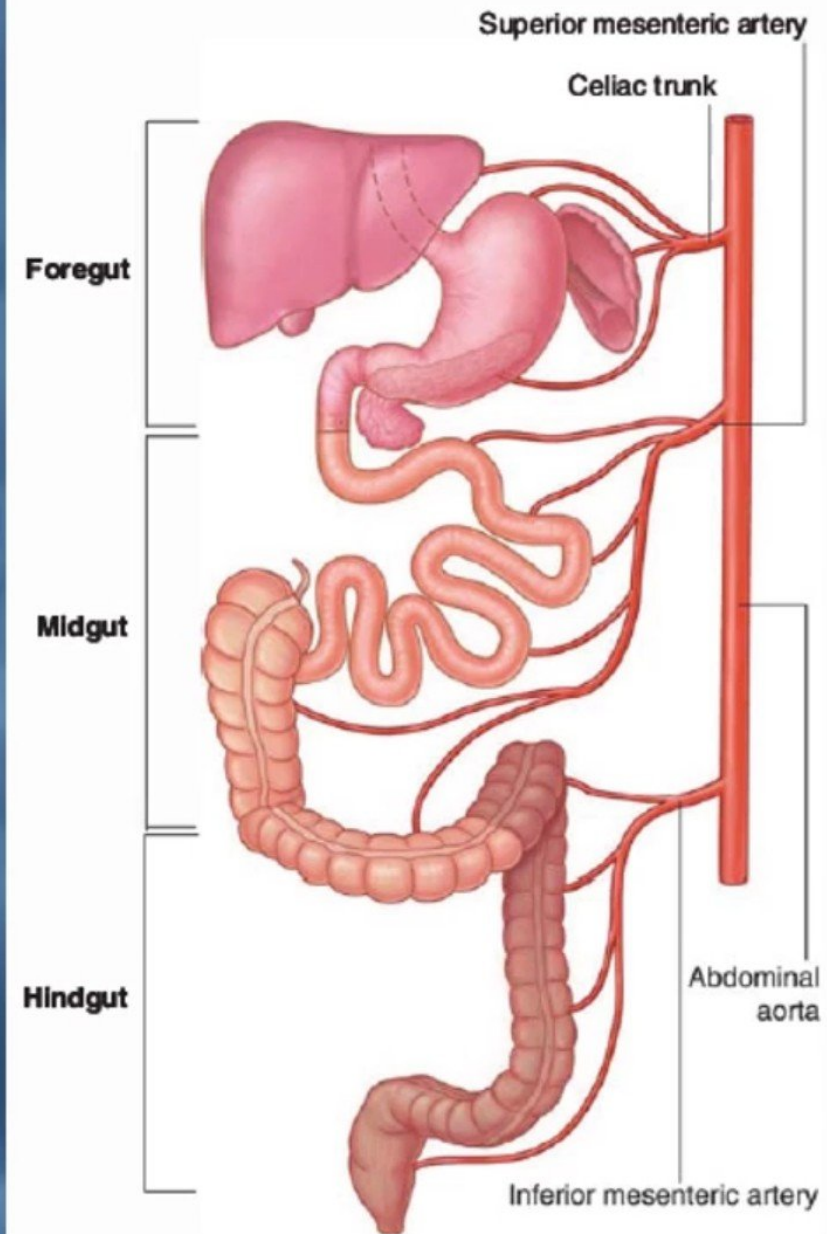


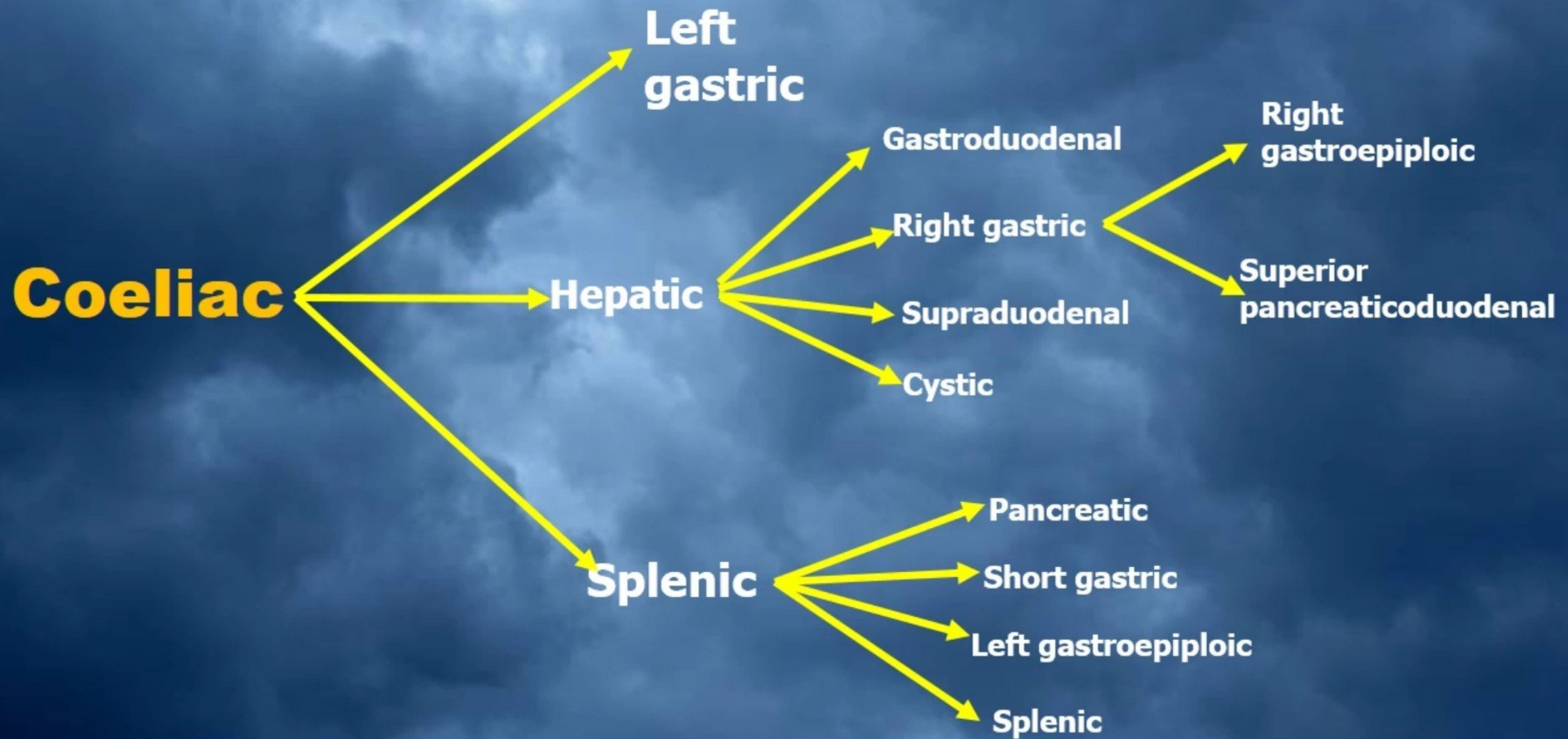
Arteries of the gut

Coeliac

Superior
mesenteric

Inferior
mesenteric





Superior Mesenteric

Inferior pancreaticoduodenal

Jejunal and Ileal

Ileocolic

Right colic

Middle colic

Inferior Mesenteric

Left colic

Sigmoid

Superior rectal

Histology of the Digestive System

Basic Histological Layers

1. Mucosa

- a. Epithelium
- b. Lamina Propria
- c. Muscularis Mucosae

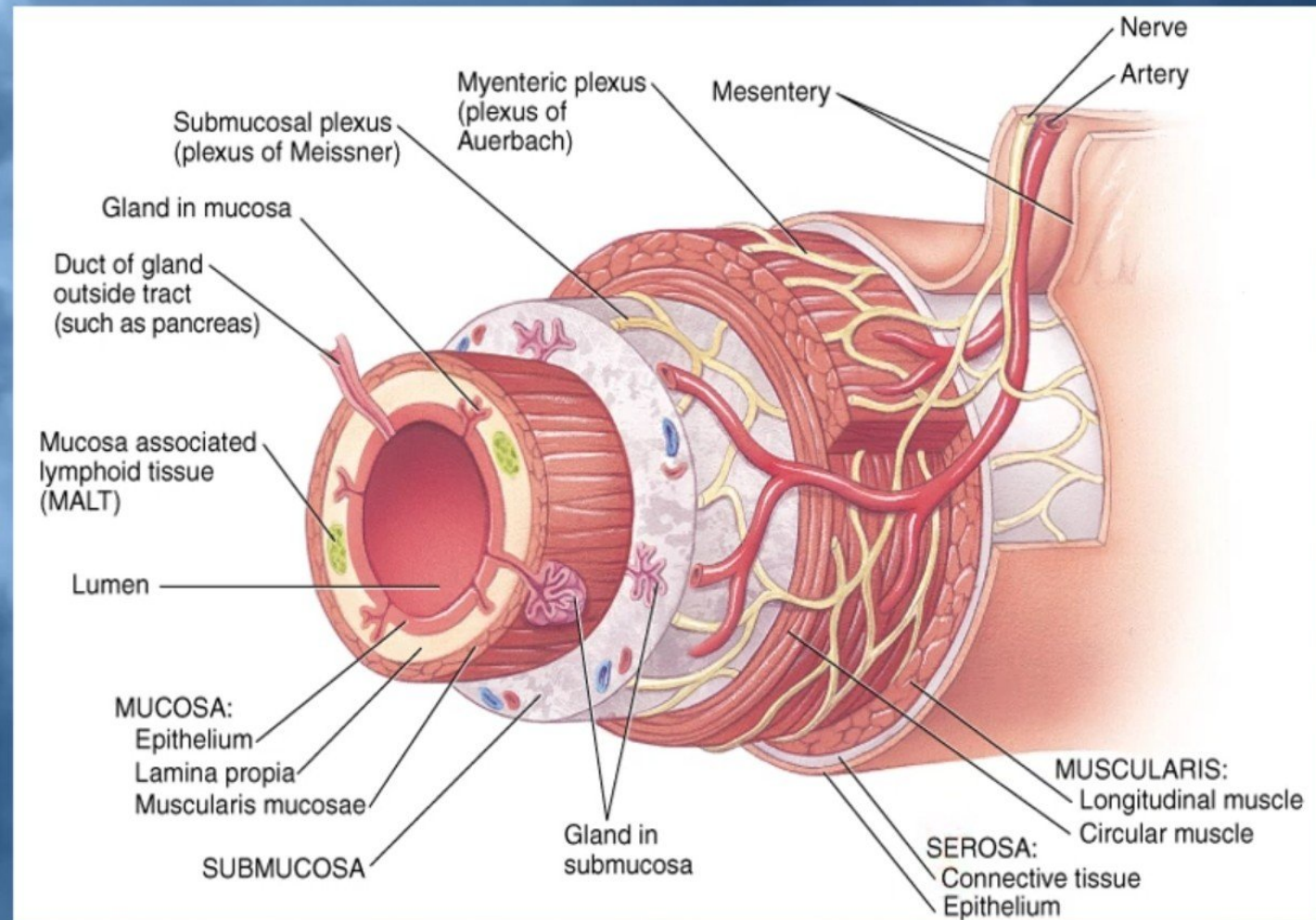
2. Submucosa

- a. Submucosal plexus
“Plexus of Meissner”

3. Muscularis

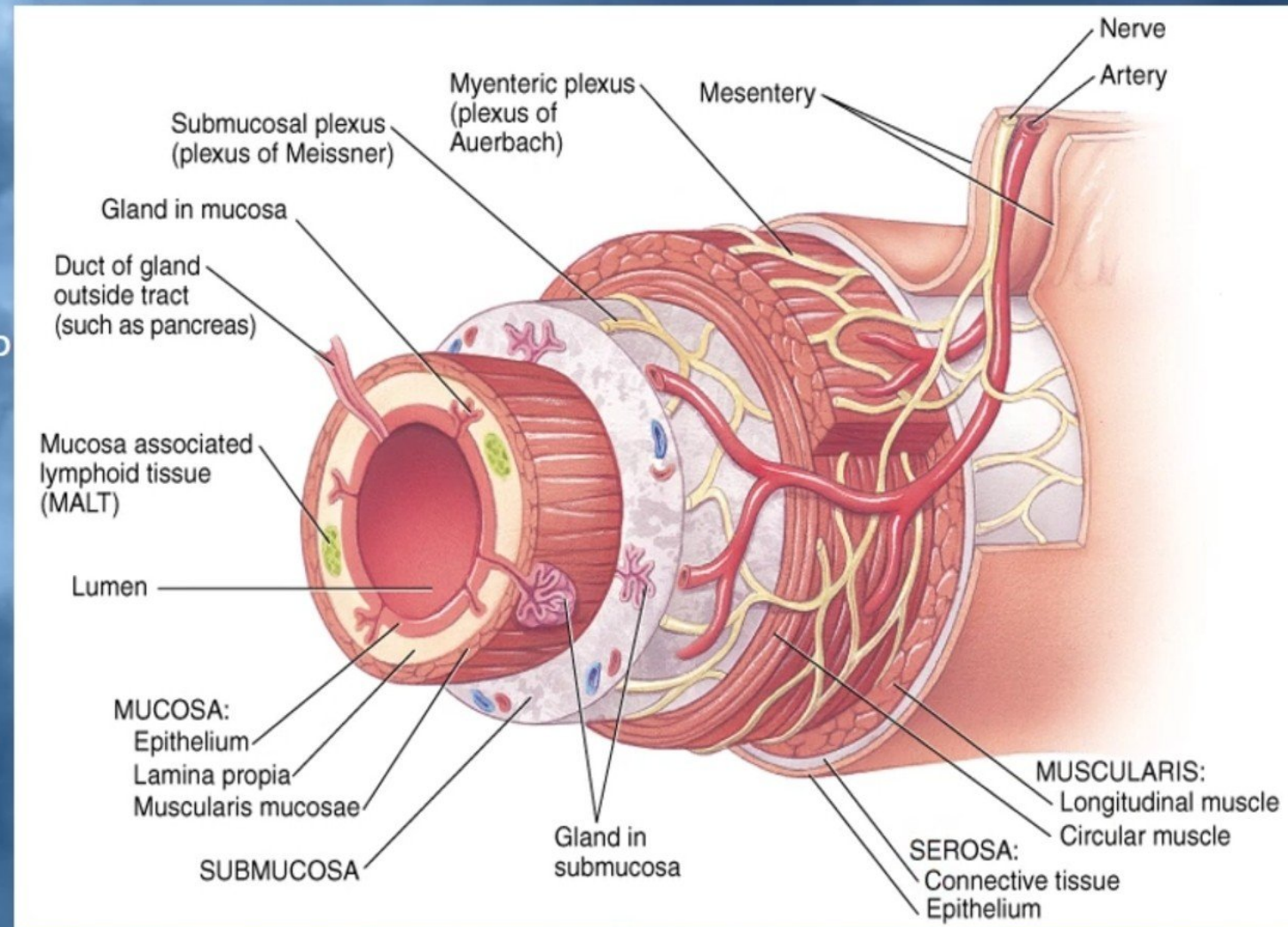
- a. Myenteric plexus
“Plexus of Auerbach”

4. Serosa



Enteric Nervous System

- The **enteric nervous system** is embedded in the lining of the gastrointestinal system, **beginning** in the **esophagus** and **extending** down to **the anus**.
- The **neurons** of the **enteric nervous system** are collected into two types of ganglia:
 - 1) Myenteric (Auerbach's) plexuses.
 - 2) Submucosal (Meissner's) plexuses.
- Myenteric plexuses are located between the **inner** and **outer layers** of the **muscularis externa**, while **submucosal plexuses** are located in the **submucosa**.



Thanks